

Technical Report

**Edge Based Hybrid System Implementation
for Long Range Safety and Healthcare IoT
Applications**

Submitted By

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in partial fulfilment of the requirements for the course of

23CSE474 - COMPUTATIONAL INTELLIGENCE

Academic Year 2025-26 Odd Semester



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

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OCTOBER 2025

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**Edge Based Hybrid System Implementation for Long Range Safety and Healthcare IoT Applications**", submitted by Yogini Aishwaryaa P T S (CB.SC.U4CSE23557) and Reenu LK (CB.SC.U4CSE23440), in fulfillment of the requirements for the course 23CSE474 – Computational Intelligence, as part of the curriculum for the Bachelor of Technology in Computer Science and Engineering, is a Bonafide record of the work carried out under our guidance and supervision at the Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore.

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List of Abbreviations

This section provides a consolidated list of all abbreviations used throughout the document, detailing their full form.

Abbreviation	Full Form
CI	Computational Intelligence
ECG	Electrocardiogram
CNN	Convolutional Neural Network
1-D CNN	One-Dimensional Convolutional Neural Network
GRU	Gated Recurrent Unit
Bi-GRU	Bidirectional Gated Recurrent Unit
ANFIS	Adaptive Neuro-Fuzzy Inference System
FIS	Fuzzy Inference System
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
DL	Deep Learning
GA	Genetic Algorithm
FL	Federated Learning
RNN	Recurrent Neural Network
LoRa	Long Range

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1 Abstract

The integration of edge computing and artificial intelligence has redefined the efficiency and responsiveness of modern healthcare IoT systems. However, traditional cloud-dependent frameworks continue to face challenges such as high latency, limited bandwidth, and data privacy concerns. This paper presents an Edge-Based Hybrid System Implementation for Long-Range Safety and Healthcare IoT Applications that leverages a combination of 1D Convolutional Neural Networks (1D-CNN), Gated Recurrent Units (GRU), Fuzzy Logic, and Genetic Algorithms (GA) to achieve intelligent and real-time ECG signal analysis directly at the edge.

The proposed framework employs a lightweight CNN-GRU hybrid model for feature extraction and temporal pattern recognition, while a Fuzzy Inference System enhances interpretability and reduces uncertainty in classification. The Genetic Algorithm optimizes model hyperparameters, ensuring efficient performance under varying computational and energy constraints. An adaptive sampling mechanism adjusts the data acquisition rate based on signal variability and patient condition. During periods of stable physiological activity, the sampling frequency is lowered to conserve power and bandwidth, while in abnormal or high-risk conditions, the sampling rate is increased to capture finer signal details for accurate diagnosis.

By processing signals locally on edge devices, the system minimizes latency and bandwidth usage, achieving fast response times and robust operation in remote healthcare environments. Experimental evaluation demonstrates that the hybrid approach improves diagnostic accuracy, inference speed, and resource utilization compared to conventional cloud-based or single-model solutions. The design ensures energy efficiency, scalability, and privacy preservation, making it suitable for continuous long-range health monitoring. Future work will explore the integration of Federated Learning (FL) for collaborative edge intelligence, further enhancing data security and global model adaptability.

Keywords: Edge Computing, Healthcare IoT, ECG Signal Processing, 1D-CNN, GRU, Fuzzy Logic, Genetic Algorithm, Federated Learning, Router, Gateway, Real-Time Monitoring, Sampling Time.

2 Introduction

2.1 Background

Edge-based systems are essential in modern healthcare and safety applications where continuous monitoring and quick response are required. Transferring all sensor data to the cloud causes high latency, bandwidth overload, and energy inefficiency. To overcome these issues, computation is moved to the edge, enabling real-time processing near the data source.

This project implements an Edge-Based Hybrid System for real-time ECG analysis and health condition classification. An Router handles signal preprocessing using bandpass, notch, and wavelet filters to remove noise, while a Gateway executes advanced Computational Intelligence (CI) algorithms.

The hybrid CI model integrates a 1D CNN for feature extraction, a GRU for temporal analysis, and Fuzzy Logic for interpretable classification. A GA optimizes model parameters to enhance accuracy and efficiency. Additionally, an adaptive sampling mechanism dynamically adjusts sampling rates based on signal activity to balance energy use and response speed.

Overall, the system achieves low latency, efficient resource use, and reliable real-time monitoring suitable for long-range IoT healthcare applications.

2.2 Motivation

In long-range healthcare and safety systems, quick detection of critical conditions is crucial. However, depending entirely on cloud servers can be problematic in real-time environments. There is a need for a smart and efficient system capable of processing data and making accurate decisions directly at the edge.

There are limitations of static sampling and fixed resource allocation in existing systems. Constant high-rate sampling wastes power, while low-rate sampling may miss vital signal changes. The proposed system integrates an adaptive sampling technique that intelligently adjusts the sampling rate according to the patient's condition, ensuring accuracy.

GA, a deep learning model used for fine-tuning, is used to automatically optimize the parameters of the CNN-GRU-Fuzzy hybrid model, allowing the system to achieve high accuracy while maintaining low latency and minimal power consumption.

Therefore, the project aims to develop a computationally intelligent edge system that performs real-time ECG analysis, adapts resource usage, and ensures reliable operation in long-range healthcare monitoring applications.

2.3 Problem Definitions

2.3.1 Purpose

The aim of this research is to address the challenges in real-time health monitoring on edge devices, focusing on reducing latency, improving classification accuracy, and handling uncertainty in physiological data. The proposed solution leverages a hybrid computational intelligence model combining 1D-CNN, GRU, and Fuzzy Logic for efficient decision-making.

2.3.2 Main Objective

To develop a hybrid computational intelligence model that integrates CNN for feature extraction, GRU for temporal pattern recognition, and Fuzzy Logic for interpreting uncertain physiological signals, enabling robust real-time health monitoring on resource-constrained edge devices.

2.3.3 Specific Goals

- Minimize latency and bandwidth usage in IoT sensor networks.
- Enhance the accuracy and reliability of health state classification.
- Incorporate fuzzy reasoning to handle uncertainty in biomedical signals.
- Optimize computational efficiency for low-power edge devices.

2.3.4 Input and Output

- **Input:** Biomedical signals (ECG) acquired from wearable IoT sensors.
- **Processing:**
 1. CNN extracts spatial features from input signals.
 2. GRU captures temporal dependencies and sequential patterns.
 3. Fuzzy Logic interprets model outputs under uncertain conditions to generate actionable health insights.
- **Output:** Classified health state (e.g., normal or abnormal) for timely intervention.

2.3.5 System Constraints

- Edge devices have limited memory and processing power.
- Communication occurs over low-bandwidth networks (LoRa, WiFi, BLE).
- Real-time response is critical, requiring decisions within milliseconds.

3 Literature Survey

3.1 Real-Time Patient-Specific ECG Classification by 1-D CNN — Kiranyaz, Ince, et al.

3.1.1 Overview of the Paper

The model emphasizes lightweight design for deployment on resource-constrained devices, enabling inference without cloud dependency. Patient-specific training improves sensitivity to individual ECG variations, enhancing detection accuracy. Experiments on standard ECG datasets demonstrate robust performance for normal and abnormal heartbeats. The study primarily extracts spatial features, without modeling long-term temporal dependencies. The compact architecture allows low-latency operation, suitable for wearable monitors. Personalized models outperform generic approaches in classification accuracy. The system mainly uses single-lead ECG signals, limiting complexity.

3.1.2 Challenges

- Designing a lightweight 1D-CNN model for real-time ECG classification on low-power edge devices.
- Adapting the network for patient-specific signal variations to improve diagnostic accuracy.
- Ensuring robust operation under noisy and dynamic ECG conditions for continuous monitoring.

3.1.3 Research Gaps

- Model Generalization: Methods to quickly adapt and generalize patient-specific models to multi-patient and heterogeneous datasets remain inadequately explored.

- **Temporal Feature Integration:** There is a need for robust and efficient exploration of long-term temporal dependencies in ECG sequences using lightweight architectures compatible with the edge.
- **Multi-Lead Applicability:** The lack of evaluation on multi-lead ECG signals represents a significant gap for comprehensive clinical monitoring systems.

3.2 Explainable Attention-based Hybrid CNN-GRU Model for Arrhythmia — Mahel et al.

3.2.1 Overview of the Paper

Mahel et al. introduced a hybrid CNN-GRU model with attention for arrhythmia detection, combining spatial and temporal feature extraction. CNN layers extract local ECG patterns, and GRU layers capture sequential dependencies. The attention mechanism highlights relevant signal segments, improving interpretability. Validated on public ECG datasets, the model outperforms standalone CNN or GRU models. The study emphasizes explainable AI, providing insight into which parts of the signal drive predictions. Offline evaluation demonstrates high classification accuracy. The trade-off between interpretability, performance, and computational complexity is highlighted. Energy-efficient or real-time deployment strategies are not explored. Overall, it provides a robust framework for interpretable hybrid deep learning models.

3.2.2 Challenges

- Combining spatial and temporal feature extraction using CNN and GRU architectures effectively.
- Developing an attention mechanism to improve model interpretability and focus on critical ECG regions.
- Managing the trade-off between accuracy, transparency, and computational complexity in hybrid networks.

3.2.3 Research Gaps

- **Efficient Attention:** Optimization methods for attention mechanisms are needed to balance interpretability with faster and more efficient inference on low-power hardware.

- **Scalable Validation:** The existing research needs extension to larger and more diverse ECG datasets to confirm scalability and performance reliability across different patient populations.
- **Edge Deployment Strategy:** Detailed evaluation and strategic implementation in real-time or resource-constrained edge environments remains an unaddressed area for this specific architecture.

3.3 Deep Learning-Based Detection of Arrhythmia Using ECG: CNN + Bi-GRU + Attention — Reshad et al., 2025

3.3.1 Overview of the Paper

Reshad et al. proposed a stacked CNN with bidirectional GRU and attention for arrhythmia detection. CNN layers extract hierarchical spatial features, Bi-GRU captures both past and future temporal dependencies, and attention highlights important ECG segments. The model demonstrates superior accuracy and sensitivity on multiple datasets. The study emphasizes capturing long-term temporal dependencies for complex arrhythmia patterns. Evaluation is primarily offline, and high computational requirements are reported. While robust, the model's performance on resource-constrained platforms or real-time scenarios is not assessed. Overall, it highlights the advantage of combining spatial, temporal, and attention-based processing for arrhythmia detection.

3.3.2 Challenges

- Capturing long-range temporal dependencies using bidirectional GRUs for complex arrhythmia patterns.
- Integrating CNN, Bi-GRU, and attention modules to enhance detection sensitivity and feature fusion.
- Maintaining training stability and efficient computation in a deep hybrid network structure.

3.3.3 Research Gaps

- **Lightweight Architectures:** Dedicated investigation of lightweight or compressed Bi-GRU and attention architectures is an inadequately studied area for maintaining accuracy while meeting edge constraints.

- **Generalizability Testing:** Comprehensive evaluation across different patient populations, including geriatric and pediatric groups, is needed to confirm the model's generalizability in clinical practice.
- **Input Optimization:** The assessment of varying input sequence lengths on model effectiveness has not been adequately addressed, which is crucial for optimizing data processing windows.

3.4 Heartbeat Sound Classification: CNN + BiGRU + Neuro-fuzzy (ANFIS) — Yadav et al., 2023

3.4.1 Overview of the Paper

CNN layers extract spatial features, Bi-GRU captures temporal dynamics, and ANFIS provides fuzzy reasoning for uncertainty handling. The system performs robustly on noisy biomedical data, outperforming standalone CNN or RNN models. The study emphasizes explainable hybrid learning, improving interpretability and reliability. However, the model is computationally heavy and evaluated offline. Overall, it demonstrates the effectiveness of combining temporal, spatial, and fuzzy reasoning in biomedical signal classification.

3.4.2 Challenges

- Achieving effective integration between deep learning and fuzzy reasoning through CNN–BiGRU–ANFIS fusion.
- Handling noisy biomedical signals while preserving feature interpretability and diagnostic reliability.
- Optimizing parameter calibration between neural and fuzzy components for balanced accuracy and efficiency.

3.4.3 Research Gaps

- **Framework Adaptation:** The adaptation of this powerful, interpretable hybrid CNN-BiGRU-ANFIS framework to analyze other biomedical signals, particularly ECG, is an inadequately studied area.
- **Complexity Reduction:** Dedicated research into architectural modifications and optimization techniques to significantly reduce the computational complexity for real-time edge deployment is required.

- **Dataset Robustness:** Despite its reported robustness, evaluation on larger and more diverse datasets remains necessary to fully validate the system's performance and prevent overfitting to specific acoustic characteristics.

4 Problem Formulation

Modern healthcare IoT systems rely heavily on cloud computing for continuous patient monitoring and analysis. While effective in theory, this dependency introduces challenges such as high latency, bandwidth overload, data privacy concerns, and inefficient energy use—all of which can delay critical decision-making. In real-time safety and healthcare applications, such delays or data losses may have serious consequences.

The main problem addressed in this work is the design of a computationally intelligent edge-based system capable of performing real-time ECG signal processing and classification locally, with minimal reliance on cloud computation. The system must efficiently manage resource allocation, sampling rate, and data transmission while maintaining high accuracy and low power consumption suitable for long-range healthcare environments.

To achieve this, the project integrates CI techniques namely 1D-CNN for spatial feature extraction, GRU for temporal sequence learning, and Fuzzy Logic for interpretable classification. A GA is employed to optimize model parameters for improved performance under limited computational resources. The system also includes an adaptive sampling mechanism that dynamically adjusts the sampling time based on ECG signal activity, balancing accuracy and energy efficiency.

4.1 System Overview

The proposed Edge-Based Hybrid System operates across three layers—Input Layer, Edge Layer, and Cloud Layer—each handling a specific stage of data flow and processing to achieve intelligent, real-time ECG analysis.

4.1.1 Input Layer

The Input Layer is responsible for data collection through ECG sensors attached to the subject. These sensors continuously capture analog physiological signals and transmit them to the edge devices. No computation is performed here; the focus is solely on accurate and continuous signal acquisition to ensure reliable input for the next stage.

4.1.2 Edge Layer

The Edge Layer performs both signal preprocessing and intelligent analysis, making it the core of the system. It consists of two key hardware components: Router: Handles signal preprocessing using bandpass, notch, and wavelet filters to remove noise and interference. This ensures that only clean, relevant data are passed to the next stage while conserving energy and reducing communication load. Gateway: Executes the Computational Intelligence model, integrating 1D-CNN, GRU, and Fuzzy Logic. The GA fine-tunes model parameters to achieve optimal accuracy and processing efficiency. An adaptive sampling mechanism dynamically modifies the sampling rate based on signal variation—reducing power consumption during stable conditions and increasing responsiveness during irregular heart activity.

This layer performs all real-time processing and decision-making at the edge, minimizing the need for cloud dependence.

4.1.3 Cloud Layer

The Cloud Layer is dedicated to data storage and management. Processed outputs and classified results from the edge are securely stored in the cloud for long-term record keeping, trend analysis, and remote access by medical personnel. No real-time computation is performed here, ensuring low latency and improved privacy while retaining historical access to health data.

5 Proposed Architecture

The Proposed Architecture provides the practical implementation framework for the formulated problem. It follows a three-tier structure—Input, Edge, and Cloud—ensuring optimized processing, minimal delay, and reliable long-range data handling.

The Input Layer serves as the sensing interface, collecting ECG data in real time. The Edge Layer acts as the computational core, where the Router performs preprocessing and the Gateway executes the hybrid CI model (1D-CNN + GRU + Fuzzy Logic). The GA refines the model parameters to enhance classification accuracy while maintaining computational efficiency. The adaptive sampling module dynamically manages signal acquisition frequency to optimize energy use and responsiveness.

Finally, the Cloud Layer stores the processed results for long-term analysis, reporting, and system updates. This division of tasks ensures that data are pro-

cessed locally for faster response, while storage and visualization remain securely managed in the cloud.

Overall, the proposed architecture provides a scalable, energy-efficient, and intelligent edge computing framework capable of reliable real-time ECG classification and long-range health monitoring.

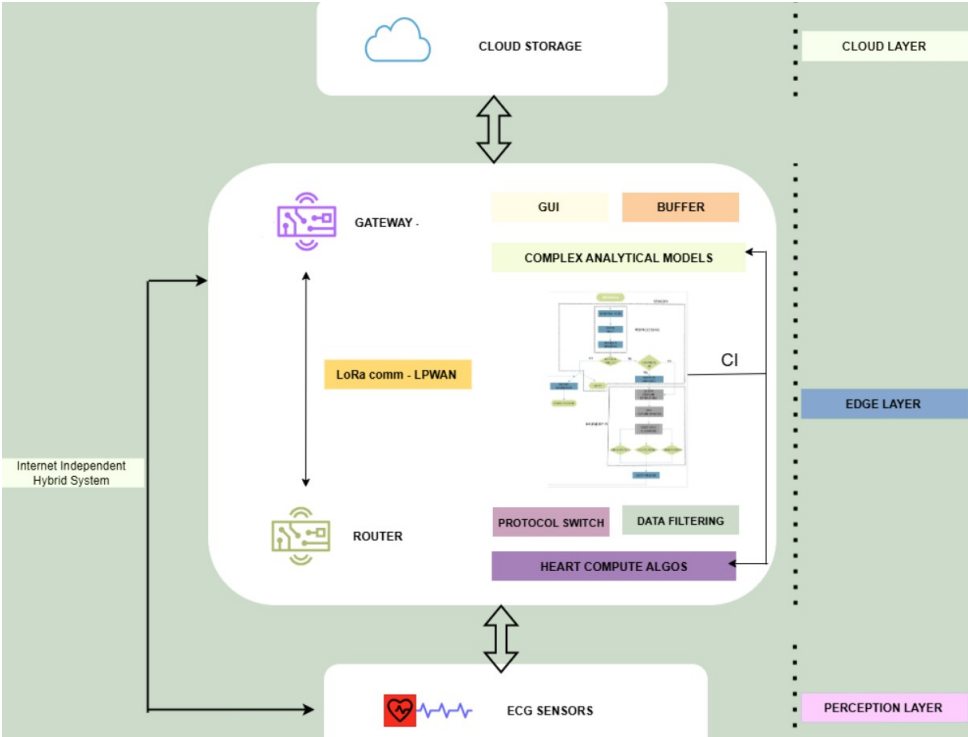


Figure 1: Architecture Diagram - both CI and Edge

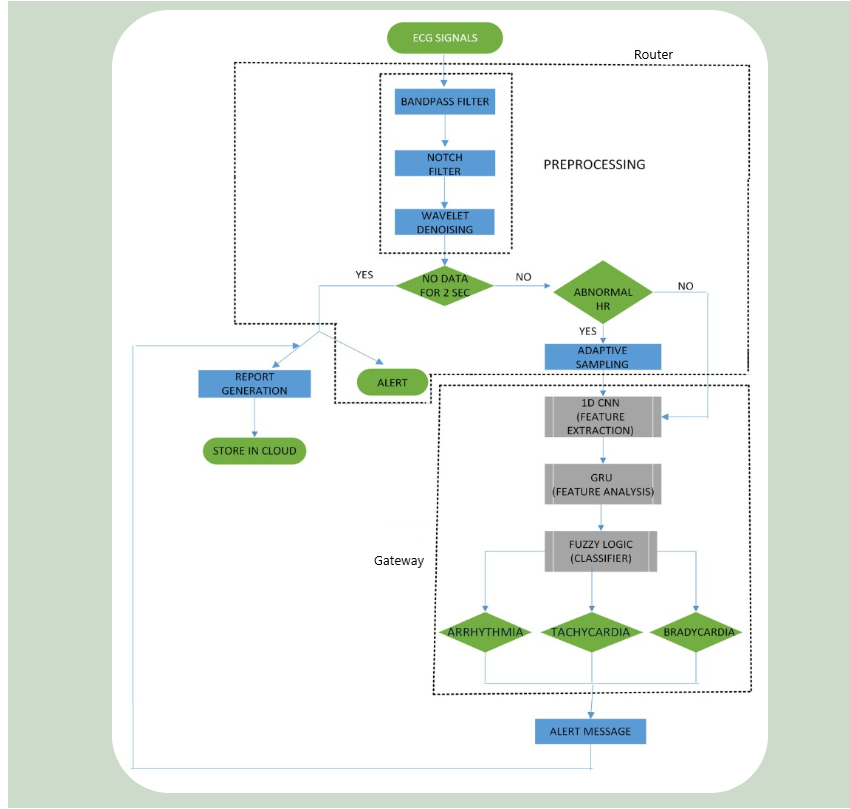


Figure 2: Architecture Diagram - only CI

6 Methodology

The proposed system follows a structured five-stage methodology to achieve accurate and real-time ECG signal classification on edge devices. Each stage is optimized for low latency, computational efficiency, and clinical reliability in long-range healthcare IoT applications. The architecture integrates traditional signal processing, deep learning, and fuzzy reasoning for robust and interpretable health assessment.

6.1 Preprocessing

Raw ECG signals are first processed to remove noise and interference introduced during acquisition. A bandpass filter(0.5–40 Hz) is used to eliminate baseline wander and high-frequency noise. A notch filter centered at 50/60 Hz suppresses power-line interference. Wavelet-based denoising further refines the signal by reducing motion artifacts and transient disturbances without distorting clinically relevant features.

This multi-stage preprocessing ensures high signal quality, essential for reliable feature extraction in subsequent stages. All these operations are implemented on the Router , allowing efficient on-board signal conditioning before

transmission to the edge node.

6.2 Adaptive Sampling

An adaptive sampling mechanism dynamically adjusts the sampling rate based on signal behavior. During stable cardiac activity, a lower sampling rate minimizes data load and power consumption. When irregularities or abnormalities are detected, the system automatically increases the sampling frequency to capture finer ECG details. This adaptive strategy enhances both energy efficiency and diagnostic precision, making the system suitable for continuous long-term monitoring.

6.3 One-Dimensional Convolutional Neural Network (1D-CNN)

The preprocessed ECG data is fed into a 1D Convolutional Neural Network for local feature extraction. Convolutional layers detect morphological patterns such as P-waves, QRS complexes, and T-waves. Pooling layers reduce dimensionality while preserving key structural information. The CNN effectively captures spatial dependencies and local signal variations, forming a compact representation of the cardiac cycle. This module performs the initial interpretation of heart signal morphology and identifies potential irregularities.

6.4 Gated Recurrent Unit (GRU) Network

The extracted CNN features are passed into a GRU network to analyze temporal dependencies between consecutive cardiac cycles. The GRU efficiently models sequential ECG dynamics to distinguish normal from abnormal rhythms. Its gating mechanism allows selective information retention, reducing memory usage compared to traditional RNNs or LSTMs. The GRU layer acts as the final deep learning stage, performing sequence-level evaluation and generating a preliminary classification output.

6.5 Fuzzy Logic Decision Layer

The final decision-making process is governed by a Fuzzy Inference System (FIS). Inputs from the CNN and GRU outputs are combined with expert-defined fuzzy rules. The FIS converts numerical probabilities into interpretable linguistic categories such as Normal, Borderline, or Abnormal. Membership functions and fuzzy reasoning handle signal uncertainty, noise, and ambiguous conditions that may not be captured by neural networks alone. This hybrid decision layer

ensures interpretability, robustness, and clinical trustworthiness of the final diagnostic output.

The proposed edge-based hybrid pipeline leverages signal processing for data refinement, adaptive sampling for efficiency, 1D-CNN and GRU for feature learning, and fuzzy logic for interpretable decision-making. Together, these modules create an intelligent, low-power, and real-time ECG monitoring system optimized for long-range healthcare IoT applications.

7 Results

The Computational Intelligence (CI) model was trained and validated using the MIT-BIH Arrhythmia dataset from PhysioNet, which contains ECG recordings from multiple patients with various cardiac conditions. The dataset was first processed to include only the required ECG classes suitable for binary classification (normal and abnormal). The class distribution was balanced, resulting in 17,327 samples after filtering.

7.1 Preprocessing

Before training, the dataset underwent several preprocessing steps to improve data quality and training efficiency. The raw ECG signals were cleaned using bandpass, notch, and wavelet filters to remove baseline drift and high-frequency noise. The cleaned signals were then segmented into uniform samples, normalized, and split into training (14,727 samples) and validation (2,600 samples) sets.

This preprocessing ensured that the input signals were consistent, reduced the model's computational load, and improved convergence speed. These same filtering and segmentation techniques were later replicated on the Router to maintain consistency between the trained model and the real-time input data.

7.2 Model Training

The hybrid CI model, combining 1D-CNN, GRU, and Fuzzy Logic, was trained on the preprocessed ECG data. The CNN extracted spatial features, while the GRU learned the temporal dependencies between ECG waveforms. The Fuzzy Logic layer converted the model's confidence levels into interpretable diagnostic decisions, such as normal or abnormal heart activity.

Training was conducted for maximum epochs using a CPU environment. The GA was used to optimize the model's hyperparameters, including learning rate

and hidden unit sizes, ensuring that the model remained lightweight for edge deployment. The training achieved a final validation accuracy of 95.65 per cent, with loss decreasing steadily over each epoch, indicating stable learning and high generalization performance.

7.3 Performane Evaluation

The model’s performance was evaluated using standard metrics such as precision, recall, and F1-score, all achieving an average of 0.96 on the validation dataset. The confusion matrix showed strong classification performance, with minimal false positives and false negatives across both classes. This demonstrates the model’s reliability for real-time ECG classification tasks.

- class 0 - Normal
- class 1 - Bradycardia
- class 2 - Tachycardia
- class 3 - Arrhythmia

Metric	Precision	Recall	F1-score
Class 2	0.98	0.93	0.96
Class 3	0.93	0.98	0.96

Table 1: Performance Evaluation

7.4 Model Optimization and Deployment

After successful training, the model was saved in the output directory as `ecg_classifier.pth`, along with a label mapping file `label_mapping.npy`. This modular saving approach allows the trained model to be reloaded and fine-tuned for future iterations.

The trained model was then compressed and optimized to ensure compatibility with edge hardware. The final lightweight version was deployed on the Gateway, which receives preprocessed data from the Router . This division ensures that the hardware load remains minimal while maintaining high classification accuracy and real-time response.

A sample fuzzy decision output confirmed that the model can interpret conditions such as “Heart Rate = 150 bpm, Confidence = 0.9 → Class 2 (Abnormal)”, validating the fuzzy-based interpretability in practical scenarios.

7.5 Summary

The results demonstrate that the proposed CI-based edge system can effectively classify ECG signals with high accuracy while remaining computationally efficient. The preprocessing pipeline, hybrid CNN–GRU–Fuzzy architecture, and GA optimization collectively contribute to a robust, low-latency model suitable for real-time, long-range healthcare IoT applications.

8 Conclusion

This work presents a hybrid edge-based system integrating 1D-CNN, GRU, and Fuzzy Logic for real-time ECG analysis in healthcare IoT applications. The architecture effectively combines traditional signal preprocessing with adaptive sampling to ensure high-quality input and efficient resource utilization on edge devices. The CNN and GRU jointly enable accurate spatial–temporal feature extraction, while the fuzzy inference layer enhances interpretability and robustness in decision-making under uncertain signal conditions.

By performing computation locally at the edge, the system achieves reduced latency, lower bandwidth usage, and improved data privacy compared to conventional cloud-based approaches. The integration of a GA for parameter optimization further refines model performance and adaptability across varying conditions.

Overall, the proposed framework demonstrates a scalable, efficient, and intelligent solution for continuous, long-range health monitoring. Future extensions include the incorporation of FL to enable collaborative training across distributed healthcare nodes, enhancing model generalization and privacy preservation.

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